Muhamad Rizki

Software Engineer | Full-Stack, Mobile & Systems Developer

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Summary

Results-driven Software Engineer and R&D Leader with 8+ years of experience engineering high-performance web, mobile, and system-level applications. Deep technical expertise in TypeScript/JavaScript (React, Next.js), Rust, C# (.NET), and React Native cross-platform architecture. Proven track record leading technology exploration, building developer tooling, and deploying resilient production services across web and mobile ecosystems.

Skills

Languages: TypeScript, JavaScript, Rust, C#, PHP, Java (Android), Swift, Kotlin, Python, SQL

Frontend: React, Next.js, Vue.js, Tailwind CSS, HTML5, CSS3, SCSS

Mobile: React Native, Native Android (Java), iOS Native Modules (Objective-C/Swift), Cross-Platform Architecture

Backend & APIs: Node.js, .NET Core / C#, PHP (Laravel, CodeIgniter), RESTful APIs, GraphQL

Databases & ORMs: PostgreSQL, MySQL, Microsoft SQL Server, SQLite, MongoDB, Prisma, Drizzle ORM, TypeORM

DevOps & Tooling: Git, GitHub Actions (CI/CD), Docker, Cloudflare, Vercel, AWS, Linux/Bash, Metro Bundler

Experience

PT. Quadrant Synergy International, Research & Development Leader
Jakarta, Indonesia
June 2022 – present
4 years 6 months

- Direct technology strategy, architectural innovation, and R&D exploration across engineering teams for enterprise web and mobile products.
- Evaluate and introduce modern tech stacks, frameworks, and developer tooling to elevate delivery speed and system maintainability.
- Architect cross-platform foundations bridging React Native, .NET Core, and distributed backend services.
- Designed internal deployment automation and release management pipelines using App Center and CI/CD workflows.
- Mentor engineers on clean architecture, design patterns, and cross-platform native module bridging.

PT. Metamesta Internasional Teknologi (Platon), Software Engineer (Contract / Remote)
Jakarta, Indonesia (Remote)
Aug 2023 – Aug 2026
3 years 1 month

- Engineered end-to-end full-stack web and mobile applications using React Native, Next.js, Node.js, and Prisma.
- Designed robust database schemas and serverless backend services deployed on Vercel with high availability.
- Collaborated across cross-functional remote teams to deliver responsive user interfaces and performant API integrations.

PT. Quadrant Synergy International, IT Full Stack Developer
Jakarta, Indonesia
Apr 2021 – June 2022
1 year 3 months

- Developed full-stack enterprise solutions utilizing C# .NET for backend APIs and React Native for client applications.
- Created a dynamic JSON schema form generation framework in Node.js and Next.js, reducing custom form development turnaround time.
- Architected relational database structures and optimized queries on Microsoft SQL Server and PostgreSQL.

PT. Quadrant Synergy International, IT Mobile Developer
Jakarta, Indonesia
Mar 2018 – Apr 2021
3 years 2 months

- Engineered cross-platform mobile apps with React Native, deploying multiple applications to Google Play Store and Apple App Store.
- Bridged native Java (Android) and Objective-C (iOS) modules for custom device capabilities, hardware integration, and performance optimization.
- Integrated offline-first local storage, push notifications, and secure authentication flows.

Central Office of LP3I, IT Mobile Developer — Jakarta, Indonesia
Feb 2017 – Mar 2018
1 year 1 month

- Developed native Android applications using Java for LP3I students and faculty to access academic and administrative services.
- Built RESTful backend endpoints using PHP CodeIgniter and MySQL to interface mobile clients with central institutional databases.

Projects

My Zurich Advisor (MZA)
Flagship enterprise mobile application for Zurich Insurance agents across Indonesia.

- Architected cross-platform React Native client with native Kotlin & Groovy modules and offline synchronization.
- Engineered dynamic product illustration workflows and digital policy onboarding for thousands of active insurance agents.

Zurich Smart (ZSmart)
Enterprise web application and actuarial calculation engine for PT Zurich Asuransi Indonesia.

- Built high-precision C# .NET calculation engine for insurance product rules, premium projections, and unit link portfolios.
- Developed full-stack React and TypeScript portal for new business underwriting and quotation generation.

QSI Enterprise Frameworks & Microservices
Foundational developer frameworks, distributed microservices, and security infrastructure at PT Quadrant Synergy International.

- Engineered C# .NET Core enterprise security framework, cryptographic token signing, and multi-tenant RBAC/ABAC authorization.
- Built distributed event-driven pub/sub messaging architecture on MassTransit and RabbitMQ, dynamic JSON Schema form engine, and administrative address indexing service.

KorinAI Platform
Generative AI multi-agent workspace with autonomous task routing, live parallel web search, and LMS orchestration.

- Built multi-session concurrent room chat and autonomous agent routing & delegation using Next.js and Vercel AI SDK.
- Engineered live parallel web search integration and rich AI chat canvas workflows.

RustO!
Pure Rust OCR Library — Fast, safe, and cross-platform on-device inference for React Native and .NET.

- Developed an optical character recognition engine based on PaddleOCR models and RTen inference runtime.
- Eliminated Python and OpenCV runtime dependencies to deliver lightweight, memory-safe OCR (~5.4 MB binary, <100ms latency).
- Published technical architecture on Medium and interactive benchmarks at byrizki.com/rusto-rs.

Paragon Logistics & Sorting System
Industrial warehouse packing and batch sorting automation system for PT Paragon Technology and Innovation.

- Developed high-throughput barcode scanning and packing-line sorting application with local daemon and AWS S3 cloud synchronization.

Berkat Karimar Mandiri (BKM) ERP
Enterprise resource planning and industrial logistics platform for commodity hauling and fleet operations.

- Engineered core weighbridge scale integration (`vol_timbangan`) and shrinkage tracking (`susut`) with automated data synchronization.
- Developed full-stack Next.js web portal and companion React Native mobile application for driver dispatch.

Cirrust
Enterprise document management system (DMS) and workflow engine at PT Quadrant Synergy International.

- Built high-volume document ingestion, secure indexing, and multi-tenant organizational workflow governance services in C# .NET and React.

JSON Eval Rs

Blazing-fast JSON Logic evaluator built in Rust with 80+ operators, comprehensive docs, and multi-language bindings.

- Engineered a pure Rust JSON Logic evaluation engine supporting 80+ standard and custom logic operators.
- Created bindings and packages for React Native, Node.js, and .NET, enabling zero-copy, sub-millisecond evaluation across platforms.
- Deployed interactive documentation and live playground at byrizki.com/jsoneval-rs.

Manulife MiRecruit

Digital agent recruitment and onboarding mobile application for PT Asuransi Jiwa Manulife Indonesia.

- Engineered candidate tracking, verification workflows, and custom native device camera bridges using React Native.

Education

Universitas MH Thamrin, Bachelor of Information Technology (S.Kom) — Jakarta, Indonesia
- Graduated with Grade B — 2018 – 2020

Politeknik LP3I, Associate's Degree in Computer & IT Administration (A.Md) — Jakarta, Indonesia
- Graduated with Grade B — 2014 – 2017

Certifications

A2 English for Developers — Dec 2025
freeCodeCamp — Credential ID: byrizki-a2efd

Career Skills in Software Development — June 2023
LinkedIn Learning — Career Management & Software Engineering